PUBLIC HEALTH

Clinical Postings | 3rd Year MBBS

Hi Sammy!
Practical public health
= saving lives daily!
This is real medicine!

Public Health | Clinical Postings | 3rd Year | MBBS

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Vaccination Technique + Cold Chain

Vaccination schedule = always asked in exams!
Cold chain = key!

Vaccination Technique — Routes + Sites

Vaccine	Route	Site	Key Points
BCG	Intradermal	Left deltoid	0.1ml, forms papule -> scar in 6-8 wks
OPV	Oral	Mouth	2 drops, no needle, booster at 16-24 mo
IPV	IM / SC	Anterolateral thigh	Fractional (0.1ml ID) or full dose IM
Hepatitis B	IM	Anterolateral thigh	Birth dose within 24hrs, 0.5ml
Pentavalent	IM	Anterolateral thigh	DPT + HepB + Hib, 6/10/14 weeks
Measles/MR	SC	Right upper arm	9 months (MR1) & 16-24 mo (MR2)
JE	SC	Left upper arm	9 months + booster at 16-24 months
Vitamin A	Oral	Mouth	9 months then every 6 months up to 5 yrs

Cold Chain — The Lifeline of Vaccines!

- Cold chain = system of storage & transport keeping vaccines at 2-8 deg C
- Equipment: ILR (Ice Lined Refrigerator), Deep Freezer (-20C), Cold Box, Vaccine Carrier
- ILR: Stores vaccines at 2-8C, open ONLY 2 times/day (morning + evening)
- VVM (Vaccine Vial Monitor): Color changes if exposed to heat — discard if dark

Freeze-Sensitive vs Heat-Sensitive Vaccines

FREEZE-SENSITIVE (never freeze!)

- DPT / Pentavalent
- Hepatitis B
- TT / Td
- IPV (liquid form)

HEAT-SENSITIVE (keep cold!)

- BCG (most sensitive!)
- Measles / MR
- OPV
- JE (live vaccine)

★ **VVM = 4 stages (1-4)**
Stage 1-2: USE vaccine. Stage 3-4: DISCARD! Inner square darker than outer = discard!

Growth chart reading = practical exam Q! Know the Z-scores!

Growth Monitoring — WHO Standards

- WHO Growth Charts: Plot weight-for-age, height-for-age, weight-for-height
- Z-score classification: Median (0), -1 SD, -2 SD (moderate), -3 SD (severe)
- MCP Card (Mother & Child Protection): Record growth at each VHND visit
- Road to Health Chart: Upward trend = healthy, flat/falling = danger!
- Weigh monthly for first year, then every 3 months up to 5 years

Nutrition Assessment — Methods

Method	Details
Anthropometry	Weight, Height, MUAC, BMI, Head circumference
Clinical	Signs: edema, hair changes, skin changes, angular stomatitis
Biochemical	Hemoglobin, serum albumin, vitamin levels, blood glucose
Dietary	24-hr recall, food frequency questionnaire, diet diary

Malnutrition Classification — Key Tables

Classification	Parameter	Moderate	Severe
Underweight	Weight-for-Age	-2 to -3 SD	< -3 SD
Stunting	Height-for-Age	-2 to -3 SD	< -3 SD
Wasting	Weight-for-Height	-2 to -3 SD	< -3 SD
SAM (MUAC)	Mid-Upper Arm Circ	115-125 mm	< 115 mm
SAM (Clinical)	Edema + wasting	Marasmus	Kwashiorkor

MUAC — Quick Traffic Light Reference

MUAC (mm)	Color Code	Action
> 135 mm	GREEN	Normal — continue growth monitoring
125-135 mm	YELLOW	At risk — nutrition counseling, follow-up
115-125 mm	ORANGE	MAM — supplementary feeding, refer
< 115 mm	RED	SAM — urgent referral to NRC/facility

IMNCI = Integrated approach to sick child! Learn the algorithm!

IMNCI — Integrated Management of Neonatal + Childhood Illness

- WHO + UNICEF strategy — for children 0-5 years at primary care level
- Three components: (1) Health worker skills (2) Health system (3) Family/community
- Assess: Danger signs, cough/difficulty breathing, diarrhea, fever, ear, nutrition
- Classify: Color-coded — PINK (urgent referral), YELLOW (treat at home), GREEN (home care)
- Danger signs: Unable to drink, persistent vomiting, convulsions, lethargic/unconscious

Health Education Methods

Method	Details & Examples
Individual	Counseling, one-on-one interview (most effective for behavior change)
Group	Group discussion, role play, demonstration (10-25 persons)
Mass	TV, radio, newspaper, posters, street plays, social media
AV Aids	Flip charts, flash cards, models, videos, projector slides
Demonstration	Hand washing technique, ORS preparation, breastfeeding position

School Health Program — Components

- Health appraisal: Medical examination at school entry + annually
- Screening: Vision (Snellen), hearing, dental, nutritional status
- Health education: Personal hygiene, nutrition, first aid, substance abuse
- School environment: Safe drinking water, clean toilets, mid-day meal
- RBSK: Rashtriya Bal Swasthya Karyakram — screening of 0-18 yrs (4 Ds)

★ RBSK = 4 Ds Screening!
Defects at birth, Diseases, Deficiencies, Development delays (including disabilities)
Mobile health teams screen children 0-18 yrs at AWCs & schools. Free referral & treatment!

Public health practicals = easy scoring! Know the tables! Go Sammy!

Most Repeated LAQs (10–12 marks)

1. Describe the National Immunization Schedule with routes & sites
2. What is cold chain? Describe equipment, maintenance & VVM
3. Describe IMNCI — components, assessment steps, classification
4. Methods of health education — classify with examples

Most Repeated SAQs (5–6 marks)

1. Growth monitoring — methods, WHO Z-scores, growth chart interpretation
2. MUAC classification for malnutrition — color coding & action
3. Nutrition assessment methods (ABCD: Anthropometry, Biochem, Clinical, Diet)
4. School health program — components and RBSK 4 Ds
5. Cold chain equipment at PHC vs District level
6. Vaccine Vial Monitor — stages and interpretation

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
BCG route of administration	Intradermal (left deltoid)
Most heat-sensitive vaccine	BCG (live, freeze-dried)
Cold chain temperature range	2-8 degrees Celsius
VVM stages to discard vaccine	Stage 3 and 4 (inner darker)
MUAC < 115 mm indicates	Severe Acute Malnutrition (SAM)
IMNCI danger signs (any one)	Can't drink, vomiting, convulsions, lethargy
RBSK stands for	Rashtriya Bal Swasthya Karyakram
RBSK screens for (4 Ds)	Defects, Diseases, Deficiencies, Delays
Best method for behavior change	Individual counseling
ORS preparation ratio	1 packet in 1 litre clean water

♥ **Every vaccine you give protects a child, Sammy!**
Public health is the silent hero of medicine. And so are you!